

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 22: Heights and Distances

Every formula and result of the chapter on one sheet

1	Angle of elevation	Angle between the horizontal and the line of sight, object above the eye
2	Angle of depression	Angle between the horizontal and the line of sight, object below the eye
3	Elevation and depression	Angle of depression of B from A = angle of elevation of A from B
4	Height and distance	$\tan \theta = \frac{\text{height}}{\text{horizontal distance}}$
5	Height and slant length	$\sin \theta = \frac{\text{height}}{\text{slant length}}$
6	Distance and slant length	$\cos \theta = \frac{\text{horizontal distance}}{\text{slant length}}$
7	tan of standard angles	$\tan 30^\circ = \frac{1}{\sqrt{3}}, \tan 45^\circ = 1, \tan 60^\circ = \sqrt{3}$
8	sin and cos of standard angles	$\sin 30^\circ = \cos 60^\circ = \frac{1}{2}; \sin 60^\circ = \cos 30^\circ = \frac{\sqrt{3}}{2}; \sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}$
9	Objects on opposite sides	Distance between them = $d_1 + d_2$
10	Objects on the same side	Distance between them = $d_1 - d_2$
11	Useful values	$\sqrt{2} \approx 1.414, \sqrt{3} \approx 1.732$